Wenfeng Zou

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Education

Jilin University in Changchun, China | B.Sc. in Electronic Information Science and Technology Sep 2023 – Present

- Average Score: 85.71/100
- Selected Courses: Analytic Geometry of Space (96), Functions of Complex Variable and Integral Transforms (94), Linear Algebra B (93), Calculus B II (92), Probability and Statistics B (86), Information Theory and Coding Theory (86)

Publication

[1] **Wenfeng Zou**. "ASVQ: What Reparameterization Is Just Enough for Efficient Codebook Learning?" Accepted by **NeurIPS 2026**. [**Sole Author**]

Research Interests

- **Representation Learning:** Self-supervised / Unsupervised Representation Learning, Vector Quantization (VQ) for Discrete Representations, Representation Collapse in VQ-VAEs (e.g., codebook collapse, dimensional collapse).
- **Efficient ML:** Neural Network Quantization, Low-Bit LLMs Training, Binary / Ternary Neural Networks.

Research Experience

Independent Researcher Oct 2025 – Present

An Empirical Study on Exploring An Efficient Reparameterization Method for Codebook Learning

- *Problem & Motivation:* Addressing codebook collapse in VQ models is crucial for efficient discrete representation learning. Recent solutions increasingly rely on expressive codebook reparameterizations, improving adaptability at the cost of additional optimization and computational overhead. This naturally raises the fundamental question of whether rotations, dense channel mixing, or complex nonlinear transformations are truly necessary for efficient codebook learning in any case.
- *Method:* Propose Adaptive-Scale Vector Quantization (ASVQ), a lightweight reparameterization method for learning the codebook. ASVQ has zero trainable reparameterization parameters, but provides a favorable trade-off between codebook expressiveness and efficiency.
- *Experiments & Performance:* Conduct both quantitative and qualitative experiments across multiple model architectures and datasets, demonstrating strong utilization and reconstruction quality. On ImageNet-1k, with 16384 codes, under the VQVAE framework, ASVQ achieves the best 25.17 PSNR and 0.74 SSIM, the highest 15144 perplexity (PPL), and raises utilization to 99.7%. On FFHQ, with the same 16,384 codes as the standard VQGAN baselines, ASVQ raises utilization to 97.8% and achieves the best perceptual quality with 2.63 rFID and 0.07 LPIPS.
- *Deliverables:* Accepted to NeurIPS 2026.

Mitigating Representation Collapse in Vector Quantization via Dirichlet Prior Matching

- *Problem & Motivation:* VQ is a cornerstone of discrete representation learning, but conventional soft-assignment optimization faces a trade-off between quantization certainty (low conditional entropy) and codebook utilization diversity (high marginal entropy). Existing entropy-regularized methods increase marginal entropy from global statistics to mitigate codebook collapse, yet often cause sample assignments to drift toward the simplex center, which raises soft conditional entropy and leaves representation-collapse risk unresolved.
- *Method:* Propose a distribution-matching constraint within the standard hard-quantization VQ-VAE framework. During training, soft assignment probabilities derived from distance scores are matched to a structured Dirichlet prior. This enforces globally balanced code usage while suppressing simplex-center-drift risk, yielding more confident assignments and helping avoid representation collapse.
- *Experiments & Performance*: Experiments show substantial improvements in discrete representation quality and competitive performance across multiple tasks. On HumanML3D, compared with vanilla baselines, the proposed method reduced rFID from 0.287 to 0.066 and raised PPL from 48 to 358. On CIFAR-10, compare with different regularization strategies, the method achieves the best reconstruction quality and the lowest conditional entropy with high PPL.
- *Deliverables:* Authored a first-author paper (in preparation).

Honors

- Third Prize, the 16th Chinese Mathematics Competitions (CMC) Dec 2024
- University-Level Third-Class Scholarship 2025–2026

Skills

- **Programming:** Python, C
- **DevOps:** Git, Linux
- **ML frameworks:** PyTorch, Hugging Face
- **Languages:** Mandarin (native), English (CET-6 540)